

Open and sustainable AI: challenges, opportunities and the road ahead in the life sciences

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Artificial intelligence (AI) has seen transformative breakthroughs in the life sciences, expanding possibilities to interpret biological information at an unprecedented capacity. To maximize return on growing investments and accelerate progress, it is urgent to address long-standing research challenges arising from the rapid adoption of AI methods. We review the erosion of trust in AI outputs driven by poor reusability and reproducibility, and highlight their impact on environmental sustainability. Furthermore, we discuss the fragmented components of the AI ecosystem and lack of guiding pathways to support open and sustainable AI model development. In response, this Perspective introduces practical open and sustainable AI recommendations mapped to over 300 ecosystem components and provides guiding implementation pathways. Our work connects researchers with relevant AI resources, facilitating the implementation of sustainable, reusable and reproducible AI. Built upon community consensus and aligned to existing efforts, these outputs will aid future policy development and structured pathways for guiding AI implementation.

The field of AI has seen a surge in breakthrough advances across scientific domains where researchers are developing AI models at an unprecedented rate, a trend directly reflected in the proliferation of AI-focused publications¹. In particular, the life sciences domain has seen a rapid adoption and integration of AI models, spurred on by an increasing volume of expertly curated biological data through the FAIR (findable, accessible, interoperable and reusable) movement² and democratized access to powerful hardware such as graphical processing units (GPUs). Additionally, landmark successes like AlphaFold 2 (ref. 3), which addressed the

decades-old challenge of protein structure prediction, are leading to a snowballing effect in AI adoption⁴. This is evidenced by AI model applications upending traditional biological computational approaches and being integrated into life science databases and tooling^{5,6}.

Despite this seemingly positive openness to adopt AI in the life sciences, there is a darker, publication-rooted AI methodology crisis^{1,7}, in which many recent AI publications are of limited reuse value due to their opaque model descriptions⁸. These failings frequently stem from poor disclosure and provenance of the data, code and model

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details, leading to challenges in reusability and reproducibility, such as inaccessible data⁹, incomplete code repositories¹⁰ or missing model information. Researchers outputting these inadequately described publications may be unaware of the importance of these elements, an issue compounded by a lack of dedicated guidance to help researchers action AI model-sharing best practices. Furthermore, insufficient recognition for extra efforts to comprehensively describe models, and a low requirement by journals for detailed AI method disclosures act as additional catalysts to the proliferation of poorly described AI publications.

To address these systemic issues, lessons can be drawn from successful research practice reforms. The FAIR data movement² is a key example of reform in the life sciences, and was in part supported in its wide application through community-developed toolkits (for example, Research Data Management Toolkit¹¹, Pistoia Alliance Toolkit¹²). These toolkit resources act as central hubs of community-maintained knowledge and can host guided implementation pathways for research best practices. In contrast, AI researchers in the life sciences currently lack such dedicated toolkits and face challenges in navigating the fragmented ecosystem of AI components such as model registries^{13,14} and metadata standards^{15,16}. Without guiding AI best-practice implementation pathways, progress toward widely available reusable and reproducible models will be impeded.

A less visible issue magnified by the low reusability and reproducibility of AI models is environmental sustainability¹⁷. Where a model cannot be reused or extensive experimentation is needed to reproduce it, wasteful resource usage occurs. This, paired with the growth in life scientists adopting more resource-intensive deep learning approaches like biological large language models (LLMs)¹⁸ and agentic AI¹⁹, will exacerbate environmental impacts if not addressed. These issues, compounded by a lack of green AI²⁰ development techniques and rare reporting of environmental impacts, demand attention to ensure accountable and sustainable AI in the life sciences.

It has, therefore, become urgent to anticipate and mitigate the growing risks associated with insufficient descriptions of AI methodologies, which spur poor reuse and reproducibility, and impact environmental sustainability. This Perspective highlights these risks and challenges from the life sciences point of view, taking into consideration existing global efforts (for example, Bridge2AI⁸, Model Openness Framework²¹, FUTURE-AI²², REFORMS²³, MLCommons²⁴, FARR-RCN²⁵). The scope of our work addresses key challenges to achieving open and sustainable AI (OSAI) model development in the life sciences through improving reuse, reproducibility and environmental sustainability. We propose an accessible set of guiding OSAI recommendations, with novelty in their mapping to a collection of over 300 AI components that can support their implementation alongside three reference implementation pathways. While acknowledging the wider scope of AI research challenges, this Perspective focuses on areas where there are available solutions to support a collective and international implementation. Therefore, we do not directly address the areas of explainable AI²⁶, trustworthy AI²⁷, ethical AI²⁸ or the wider set of topics within responsible AI²⁹, which raise complex socio-ethical considerations and often require more nuanced approaches. Our goal is to guide and connect researchers with solutions to enhance sustainable and collaborative AI-based research, accelerating life science discovery.

Current challenges

Reusability of AI models

Despite the rapid growth in publications of new life science AI methods¹, they face a major lack of reuse, such as through adaptation and redeployment. Even within the subset of life science models that do see reuse, there are obstacles to their full exploitation. A notable example is AlphaFold 2 (ref. 3), which was missing crucial model information. This impeded its full reuse potential, which necessitated a more open community reimplementations, OpenFold³⁰, that shared all model-relevant

code and data. The barriers to reuse of AI models in the life sciences stem from a diverse number of reasons, such as narrow task applicability, the academic requirement for novelty, inequitable resource access, and the lack of explainability and trustworthiness of deep learning-based models. While not all these aspects impacting reuse can be addressed, there are a number of more practical issues that can be addressed through researcher best-practice reform (Fig. 1). These include absent or restrictive licensing, insufficient metadata and low rates of dedicated AI registry usage.

There are many existing community-developed recommendations and best practices that can be leveraged to improve AI model publishing, such as the DOME Recommendations¹, TRIPOD³¹, REFORMS²³ and Bridge2AI⁸. Additionally, there are several life science sub-domain checklists, such as the checklist for AI in bioimaging by Schmied et al.³² and CONSORT-AI³³ and SPIRIT-AI³⁴, designed for AI use in clinical trials. These resources can aid a researcher to correctly share all relevant model and dataset information. Unfortunately, many of these face low adoption among researchers, and this issue is compounded by the loose or lack of journal mandates for AI-specific method disclosures in publications. For publishers, their inclusions for AI model-specific software and data are generally loose beyond standard life science database and public software repository links such as GitHub or GitLab. During peer review, dataset and code repository links a researcher provides can often go ignored, with a reviewer instead solely focusing on the publication text. The current emphasis in research publishing involves text-based descriptions taking priority over a comprehensive sharing of the methods for real-world reuse. This leaves the practical considerations for supporting AI model reuse as an afterthought, if they are addressed at all. There are also few motivating incentives to encourage a researcher to take these extra steps despite their value to scientific advancements.

Researchers are also currently faced with too many options and a functionally diverse AI ecosystem of repositories to document and store their AI datasets, code and models. These include AI domain-agnostic solutions (for example, Kaggle³⁵, Hugging Face³⁶), life science-specific resources (for example, Kipoi¹⁴, Biolmage Model Zoo¹³, Ersilia Model Hub³⁷) or even trusted non-AI platforms (for example, Zenodo³⁸, in structured RO-Crates³⁹). This choice forces consideration of the diverse feature sets, user bases, long-term data retention policies and trustworthiness of different platforms. As a result, this has often led to inconsistent sharing, poor findability and models not being deposited in a reusable manner beyond basic raw file availability. Furthermore, a lack of trust in a potentially suitable method, due to insufficient information to fully assess it, often hinders reuse and encourages the development of new models rather than reimplementing potentially suitable ones. This can be overcome by addressing the failings of proper technical sharing of AI data, code, models and their documentation covering important aspects such as their dataset availability and provenance³⁹.

FAIR for AI initiatives⁴⁰ are looking to support reusability globally across both academia and industry, with organizations like the Research Data Alliance (RDA) FAIR4ML Interest Group¹⁶ developing model metadata (FAIR4ML schema)⁴¹, Machine Learning Commons (MLCommons)²⁴ focusing on metadata for model datasets (Croissant¹⁵) and the Pistoia Alliance⁴² addressing AI data readiness⁴³ (DataFAIRy project⁴⁴). However, despite this initial work, widely available and truly reusable AI models are still not yet prevalent. This is due to a multitude of reasons including the rapid increase in AI adoption outpacing best-practice development and, critically, a lack of targeted top-down best-practice mandates from funders, policymakers and journals.

Reproducibility of AI models

Reproducibility is a multifaceted concept and in the context of AI research has various interpretations⁴⁵. We describe the challenge of AI model reproducibility in relation to the reuse of a method's available

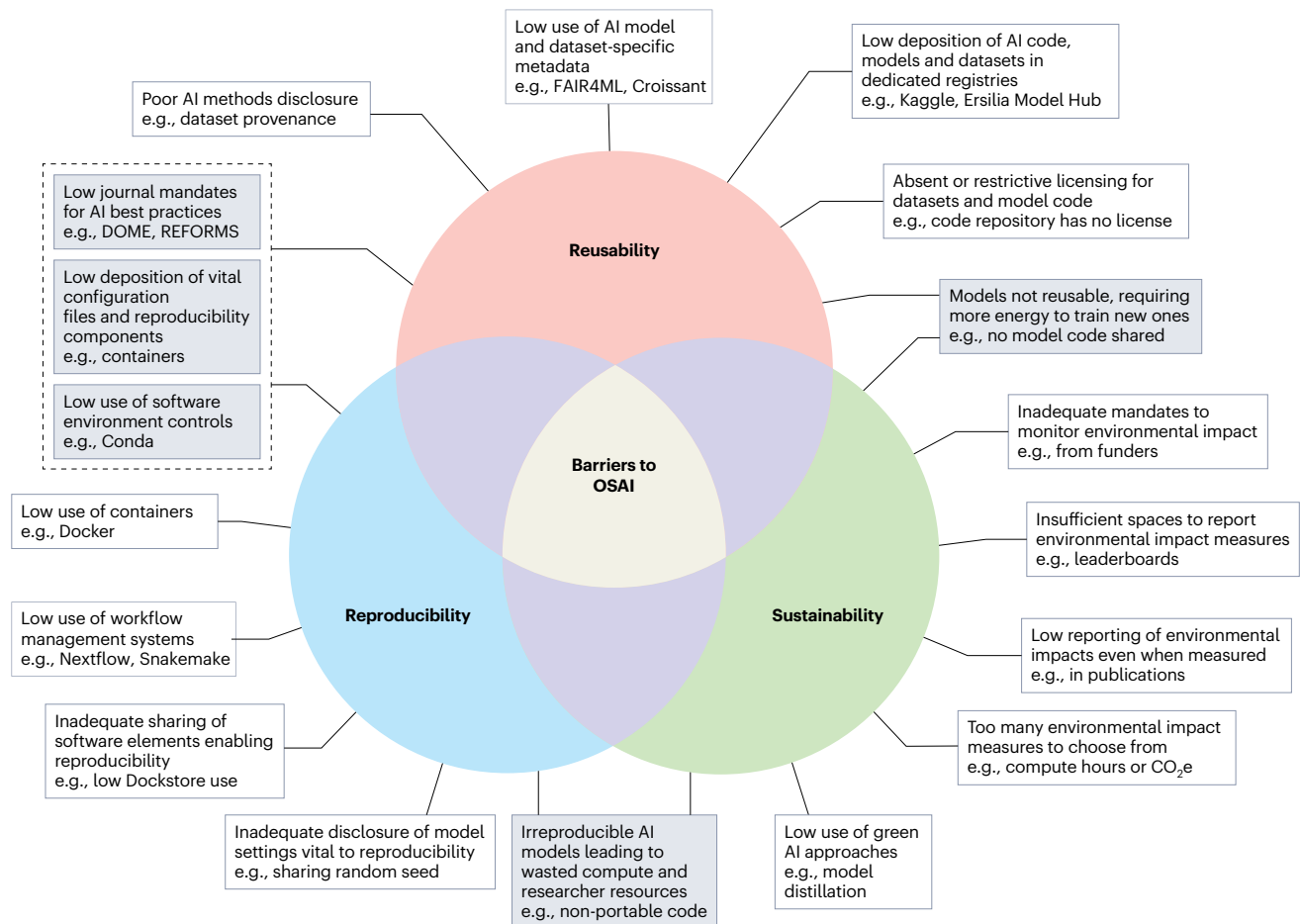


Fig. 1 | Barriers to OSAI in the life sciences. Three interlinked challenge areas of reusability, reproducibility and sustainability that must be overcome by researchers to accelerate open and environmentally sustainable AI model

development in the life sciences. Challenge areas have their key issues highlighted with examples, as well as their intersections (gray). Created in BioRender; Farrell, G. <https://BioRender.com/2rSoup5> (2026).

data and code to: facilitate a reproduction of an AI model to enhance understanding; and reproduce consistent model benchmarking results to build trust in reported performances. These are vital to extract learnings and validate previous findings from promising research to build upon with future work⁴⁶. Highlighting the widespread difficulty of AI model reproducibility in the life sciences, BioModelsML attempted to curate a collection of reproducible models from public sources but faced many time-consuming barriers during their efforts⁴⁷. Even in cases where all relevant information toward generating an AI model is openly available and clearly disclosed, software environment requirements can often be prohibitively time-consuming and impractical to replicate. Therefore, an AI model can be transparently described, but practically not reproducible due to poor portability. As such, the adoption of solutions that can support standardized reproducibility of AI methods across systems requires a broader uptake to address the reproducibility crisis facing researchers.

On the software side, several well-established resources exist to facilitate AI model portability and computational reproducibility. Many of these solutions span the areas of software containerization (for example, Docker⁴⁸), environment control (for example, Conda⁴⁹) and workflow management (for example, Nextflow⁵⁰, Snakemake⁵¹, Galaxy⁵²). These solutions supporting portable deployment across systems, provide simpler reproductions of complex computational runs with far fewer overheads⁵³. While these solutions help reduce the barriers to life science computing reproducibility, their prevalence and use for AI models is still variable among publications⁴⁷. Even though many of these reproducibility-enabling tools cover AI needs, many

AI methods only share a disjoint repository of scripts with limited instructions to easily reproduce their entire runs. Therefore, AI model reproducibility is time-consuming due to the infrequent and inconsistent application of tools capable of orchestrating complex model generation runs. Reasons for their underutilization may include the time and training needed to leverage them, as well as a misunderstanding by researchers of the overheads involved to reproduce an AI model across diverse computational environments.

Beyond software approaches for ensuring reproducibility of AI methods, the aspect of diverse physical hardware must also be considered. Computational infrastructure requirements can range drastically in developing and deploying AI models, from use of a simple laptop CPU or Google Colab⁵⁴ instance to requiring a cluster using hundreds of GPUs. Consequently, correct documentation of computing infrastructure used and configuration file sharing is important for reproducibility⁴⁵ but often lacking in life science AI model sharing. This oversight means that attempting to run and reproduce an AI method may be impossible, even when access to the required computing infrastructure is not a limitation. Moreover, researchers can face prohibitive costs when using commercial cloud solutions in attempts to reproduce an AI model where the exact hardware needs of the original model were not openly disclosed. This creates unforeseen access challenges for researchers wishing to reproduce and build upon existing AI methods. Therefore, it is crucial for researchers to closely consider transparently documenting all infrastructure and related configuration requirements in addition to taking software measures to provide portable reproducibility.

Sustainability of AI models

The challenges of poor AI method reusability and reproducibility directly impact environmental sustainability due to the need to repeat heavy energy expenditures, such as retraining models, which increases carbon emissions¹⁷. This 'red AI'²⁰ issue is compounded by the escalating resource needs of modern AI systems, particularly deep learning models. Their adoption by a wider number of life scientists intensifies the growing energy demands⁵⁵ and raises responsible AI concerns about the Global North's disproportionate environmental impact⁵⁶. Furthermore, where models are not made available and preserved, this adds to irresponsible AI energy waste. Therefore, environmental sustainability of AI systems requires accountable model development in ways that minimize negative environmental impacts while maximizing societal outcomes, a principle reflected in initiatives like the European Green Deal⁵⁷, which closely considers AI impacts.

Despite these red AI concerns, there is not yet a prominent focus in life science research on mitigating it through the adoption of green AI⁵⁸ approaches. This is partly because applying AI to biological data is already an inherently challenging and interdisciplinary task, often leaving less pressing green AI considerations like measuring and reporting environmental sustainability impacts a lower priority. The implementation and routine reporting of energy usage metrics, especially outside large-scale high-performance computing (HPC) sites, is not standard practice. This lack of monitoring and reporting often stems from insufficient requirements by organizations or funders, as AI sustainability is frequently a less visible concern. While individual projects may seem to have a small footprint, the collective global growth in the adoption of energy-intensive deep learning represents a largely unmonitored aggregate impact, necessitating better tracking. Even where large HPC centers track and report their overall consumption (for example, EMBL-EBI energy consumption report⁵⁹), environmental impact accounting attributable to specific AI model development often remains unaddressed at the individual lab and researcher level.

Compounding these issues, even when individual researchers are conscious of AI sustainability impacts, they may find it difficult to know what to measure, in what format and where to report it. Consequently, they may face challenges in identifying and understanding the limitations of common sustainability measures like CO₂ equivalent or energy consumption, each varying in accuracy and applicability (Table 1). This situation is exacerbated by a lack of accessible tooling designed for life science researchers to correctly plan, track and measure the sustainability impact of their AI methods. Furthermore, for reporting measures there is an absence of journal requirements and reporting templates.

Beyond measuring and reporting, a critical implementation gap exists in the life sciences concerning the adoption of methods to reduce AI's environmental impact, such as through green AI model development. Many approaches exist, such as Frugal AI⁶⁰, which considers energy-efficient model complexity reduction, and green AutoML techniques⁶¹ for optimizing architecture while minimizing computational cost. However, such techniques remain underutilized, leading to inefficient resource use.

Overall, while wider responsible AI governance is visible in the life sciences, often with a strong ethical-, legal- and societal-implications⁶² emphasis on human data privacy (for example, GRAIMatter highlighting AI privacy leakage risks⁶³), specific mandates for AI environmental sustainability are frequently overlooked when not driven by institutional and governing bodies. A possible factor in this context may include the ethical dilemma of attempting to balance conscious energy use against vital research often justified for societal good, such as cancer studies or drug discovery. This highlights the need to carefully manage the relation between important research freedom alongside mitigating environmental sustainability concerns. Encouragingly, initiatives like the FUTURE-AI consortium²² and the AHEAD project⁶⁴ are beginning to develop broader assessment frameworks applicable to AI in the life sciences that include technical, clinical, socio-ethical

and legal dimensions, which may pave the way for more holistic environmental sustainability evaluations. Furthermore, emerging project initiatives considering green AI (for example, in the ELIXIR STEERS project⁶⁵, SustAI nML EU project⁶⁶) or community-driven sustainable software accreditations (for example, Green DiSC Certification⁶⁷) signal a positive direction for integrating environmentally conscious solutions to meet urgent green AI needs in the life sciences.

OSAI: recommendations

In direct response to the three pressing and interlinked challenges of reusability, reproducibility and environmental sustainability, we propose nine practical and achievable OSAI recommendations that life science researchers should implement. These recommendations are structured in three tiers of implementation priority and designed to address the most overlooked but impactful considerations for creating OSAI-compliant research outputs. We recognize the preexisting body of work to guide researchers leveraging AI, such as Bridge2AI⁸ for biomedical research AI data readiness and the domain-agnostic REFORMS²³ AI disclosure checklist. However, our recommendations offer a targeted and implementation-oriented approach for the broader life sciences, specifically addressing the urgent challenges concerning both AI models and their datasets that are prevalent across the entire domain. We provide novelty with our recommendations through their support for application with a mapping back to a non-prescriptive collection of over 300 AI ecosystem components, such as AI-specific metadata, registries and tools (<https://osai.dome.ml.org/ai-ecosystem/>). This ensures researchers are both informed of the best-practice recommendations and linked with the existing solutions to accomplish OSAI.

Reusability

R1: Generate and share standardized AI metadata for models and datasets. Standardized metadata are essential to aid both human and machine understanding of AI models and their datasets for reuse. AI metadata adoption enables programmatic findability, supports interoperability and ensures correct disclosure of key model and dataset attributes. We recommend generating and exposing metadata for both AI models and their respective datasets (Box 1). For model annotation, a possible standard includes the recently launched FAIR4ML schema⁴¹ from the RDA FAIR4ML Interest Group¹⁶. For AI dataset annotation, the Croissant¹⁵ standard from MLCommons²⁴ is a suitable example due to its layered approach and domain extensibility (for example, GeoCroissant for geospatial AI datasets⁶⁸). Both are open, community-driven standards and cover diverse metadata fields such as key information like licensing and provenance. Other AI metadata standards also exist, such as Model Cards⁶⁹ and Dataset Cards⁷⁰. Additional considerations include leveraging dedicated AI ontologies like Machine Learning SAILor Ontology (MLSO)⁷¹ to help standardize metadata vocabulary.

R2: Leverage AI registries as central hubs for sharing and discovering reusable AI models and datasets. There is a growing number of AI model and dataset registries, with over 30 identified in our AI ecosystem landscaping. They support depositing, registering and sharing these valuable research objects for community reuse. While choosing the most suitable one can be difficult, a wider adoption by researchers will enhance visibility, aid co-development and functionality convergence to serve life science needs. The primary choice consideration relates to choosing either suitable life science-specific registries (for example, BioImage Model Zoo¹³ for bioimaging, Kipoj¹⁴ for genomics) or domain-agnostic registries (for example, Hugging Face³⁶, Kaggle³⁵). Additional considerations supporting this choice include the AI registry provider's commitment to open science, long-term data preservation, and support for exposing standardized model and dataset metadata.

Table 1 | Metrics a researcher could use to measure AI model computation and energy usage to infer environmental impact

Measure	Description	Measurement limitation	Unit
CO ₂ e	Estimated greenhouse gas emissions associated with the energy consumed, based on grid carbon intensity.	An estimate, highly dependent on location, time and accuracy of the energy measurement and carbon intensity factor used. Does not include hardware's embodied carbon.	Kilograms CO ₂ e Tonnes CO ₂ e
Energy consumption	Total electrical energy consumed by the hardware during the computation. Direct measure of resource use.	Can be difficult to measure accurately without dedicated power meters. Software estimates may be inaccurate or incomplete (do not account for RAM and cooling, for example).	Watt-hours Joules
Floating-point operations (FLOPs)	Count of arithmetic operations on floating-point numbers; proxy for algorithmic complexity.	Hardware-independent; does not directly measure time or energy. Ignores data movement costs (memory input output). Efficiency (FLOPs/watt or FLOPs/second) varies vastly across hardware.	FLOPs Giga-FLOPs
GPU- or CPU-hours	Total time-specific computing units (GPU cores, CPU cores) that were allocated or actively used for the task.	Does not account for varying utilization levels (idle versus 100%). Different hardware has vastly different power consumption per hour. Poor indicator of actual energy used.	Hours Core-hours (for example, GPU-hours)
Power draw	Instantaneous rate at which electrical energy is consumed by the hardware components.	Fluctuates substantially during computation. Average power needed for energy calculation. Thermal design power is often a limit, not actual usage. Software estimates vary in accuracy.	Watts Kilowatts
Total runtime	Wall-clock time elapsed from the start to the end of the computation process.	Highly dependent on specific hardware used. Does not distinguish computation versus idle or wait time. Does not directly measure energy or computational work performed.	Seconds Minutes Hours

BOX 1

Recommendations to aid AI model reuse

Reusability			
No.	Recommendation	Explanation	OSAI priority
R1	Generate and share standardized AI metadata for models and datasets	Adopt and share standardized metadata using community-driven schemas for both datasets and models to enhance machine readability, findability and interoperability for reuse.	Essential
R2	Leverage AI registries as central hubs for sharing and discovering reusable AI models and datasets	Deposit and register models and datasets in established AI registries, considering choice between domain-specific or general-purpose options to maximize visibility and accessibility for the research community.	Required
R3	Host, promote and share training and guidance on correct deposition of AI research objects for reuse	Develop and disseminate training materials on best practices for AI model and dataset deposition, utilizing dedicated training resource platforms to build community capacity and ensure proper sharing for reuse.	Recommended

R3: Host, promote and share training and guidance on correct deposition of AI research objects for reuse. There are many evolving AI model deposition considerations, such as metadata, hosting registries and related checklists, making it difficult for researchers to keep up with best practices to support AI model and dataset sharing. Therefore, it is crucial that research groups and institutions begin to develop, host and promote training and guidance on correct

AI model deposition. This will aid researchers to comprehensively share models and maximize reuse value. While AI model deposition training material examples are limited, there are institutional model deposition guidance examples, such as the Swedish SciLifeLab lightweight and nationally focused guidance for model sharing⁷², which reuses existing frameworks such as the Model Openness Framework²¹. Furthermore, to democratize the availability of these materials, we recommend hosting them in existing open infrastructure like training material hosting resources (for example, Glittr⁷³, Data Carpentry⁷⁴) or Turing Way⁷⁵ modeled toolkits (for example, RDMkit¹¹). This will be essential to democratize model-sharing knowledge and maximize reusability.

Reproducibility

R4: Ensure transparent disclosure, clear documentation and sharing of all model-relevant information. Transparent reporting of all model-relevant information is crucial to facilitate understanding and reproducibility, even if complete computational reproducibility is challenging. This includes sharing comprehensive details on model architecture and weights, preprocessing steps, model architecture, training procedures, datasets, evaluation metrics and, ideally, complemented by access to relevant reproducibility files (for example, ONNX⁷⁶). Such information should be readily available across a publication's methods section, supplementary materials or alongside code in repositories (Box 2). These disclosures should be guided by the use of comprehensive checklists (for example, Model Openness Framework²¹, REFORMS²³) and, where possible, supported by their hosting in reporting platforms like the DOME Registry⁷⁷, which provides structured information access and persistent identifiers for this information. Ultimately, transparent disclosure is crucial for ensuring clear access to information to aid reproducibility.

R5: Provide portable code and reproducible environments to facilitate smooth reuse. To enable practical reproduction of models from training to inference across diverse computational systems, researchers should leverage readily available components. These include using software containers (for example, Docker⁴⁸, Apptainer⁷⁸), environment control (for example, Conda⁴⁹) and workflow management systems (for example, Nextflow⁵⁰, Snakemake⁵¹). These should then be made available for use through open platforms (for example, DockerHub⁷⁹, Dockstore⁸⁰). While simpler methods like online and interactively

BOX 2

Recommendations to aid AI model reproducibility

Reproducibility			
No.	Recommendation	Explanation	OSAI priority
R4	Ensure transparent disclosure, clear documentation and sharing of all model-relevant information	Ensure transparent reporting by comprehensively documenting all model-relevant information, including data, methods and configurations, guided by community disclosure checklists and reporting platforms.	Essential
R5	Provide portable code and reproducible environments to facilitate smooth reuse	Facilitate practical model execution by providing portable code within reproducible environments using software containerization, environment management tools and workflow systems.	Required
R6	Use standardized AI-ready datasets and benchmarking evaluation protocols to facilitate reproducible model comparisons	Promote model comparability and validate performance claims by using standardized benchmark datasets and community-agreed evaluation protocols, often accessible via public benchmarking platforms.	Recommended

hosted scripts (for example, Google Colab⁵⁴, Code Ocean Capsules⁸¹) can be useful, sharing dedicated reproducibility components within well-annotated code repositories alongside all necessary configuration files represents best practice. Choice of tooling should consider longevity and support for components used to ensure long-term reproducibility.

R6: Use standardized AI-ready datasets and benchmarking evaluation protocols to facilitate reproducible model comparisons.

To ensure the validity of model performance claims and enable reproducible comparisons, the use of standardized evaluation benchmark datasets and protocols across models is required. AI-ready datasets as defined by the Open Data Institute's 'Framework for AI-ready data'⁴³, are those that are structured and prepared for direct use by AI models across areas such as technical optimization and adherence to standards. They are key to facilitating reproducible and comparable evaluations. Additionally, rigor in standardizing model evaluation protocols is vital for transparently reporting model performances. Community-driven efforts, such as challenges like CASP⁸² for protein structure prediction or wider collection of DREAM challenges⁸³ (for example, drug sensitivity prediction) are core to agreeing and applying standardized evaluation protocols. These efforts in turn drive scientific discovery and the development of more trustworthy and reliable model assessments. Platforms like OpenEBench⁸⁴ (life science specific) and Kaggle competitions³⁵ (domain agnostic) can aid in discovering and hosting benchmarking challenge results and information on the standardized datasets, and evaluation protocols enabling reliable comparisons.

Sustainability

R7: Implement green AI model development techniques. The diverse set of green AI model development techniques can greatly reduce energy consumption during model training and boost downstream inference efficiency (Box 3). Methods such as warmstarting⁸⁵, pruning⁸⁶, quantization⁸⁷, green auto-ML⁸⁸, knowledge distillation⁸⁸ and transfer learning⁸⁹ can substantially cut computing needs. The case of the DeepSeek-R1 model⁹⁰ exemplifies how optimizing development efficiencies can achieve more with less hardware, a practice termed 'frugal AI'⁶⁰. This demonstrated that a greater use of computation is not always synonymous with better results. Resources like green software patterns⁹¹ from the Green Software Foundation⁹² can guide researchers in applying these energy-efficient practices throughout the model development cycle.

R8: Choose and optimize hardware to reduce environmental impact. The optimization and choice of hardware, relating to microchips and location, critically impacts AI sustainability. For on-premises computing, where possible, researchers should select energy-efficient modern hardware, optimize energy settings and use chips optimized for specific tasks (for example, GPUs for deep learning). With cloud computing, options are expanded to make use of data centers in green energy-powered zones⁵⁵, scheduling workloads for optimal green energy availability. However, work with sensitive biological data and legal frameworks may limit such choices. For large-scale AI projects, considering Top500 Green List facilities⁹³ is also recommended as they account for the full carbon life cycle in their design and operation. Finally, leveraging available

BOX 3

Recommendations to aid AI model sustainability

Sustainability			
No.	Recommendation	Explanation	OSAI priority
R7	Implement green AI model development techniques	Reduce energy consumption by implementing green AI techniques (for example, pruning, transfer learning) and adopting frugal AI principles throughout the model development life cycle, guided by established best practices.	Essential
R8	Choose and optimize hardware to reduce environmental impact	Minimize environmental footprint by selecting energy-efficient hardware, optimizing chip usage for specific tasks and considering green energy sources or sustainably designed data centers for computations.	Required
R9	Measure and report AI model environmental impact	Consistently measure and report key environmental impact metrics (for example, energy consumption, carbon emissions, runtime) using suitable tooling, and disclose these figures in publications or on dedicated reporting platforms.	Recommended

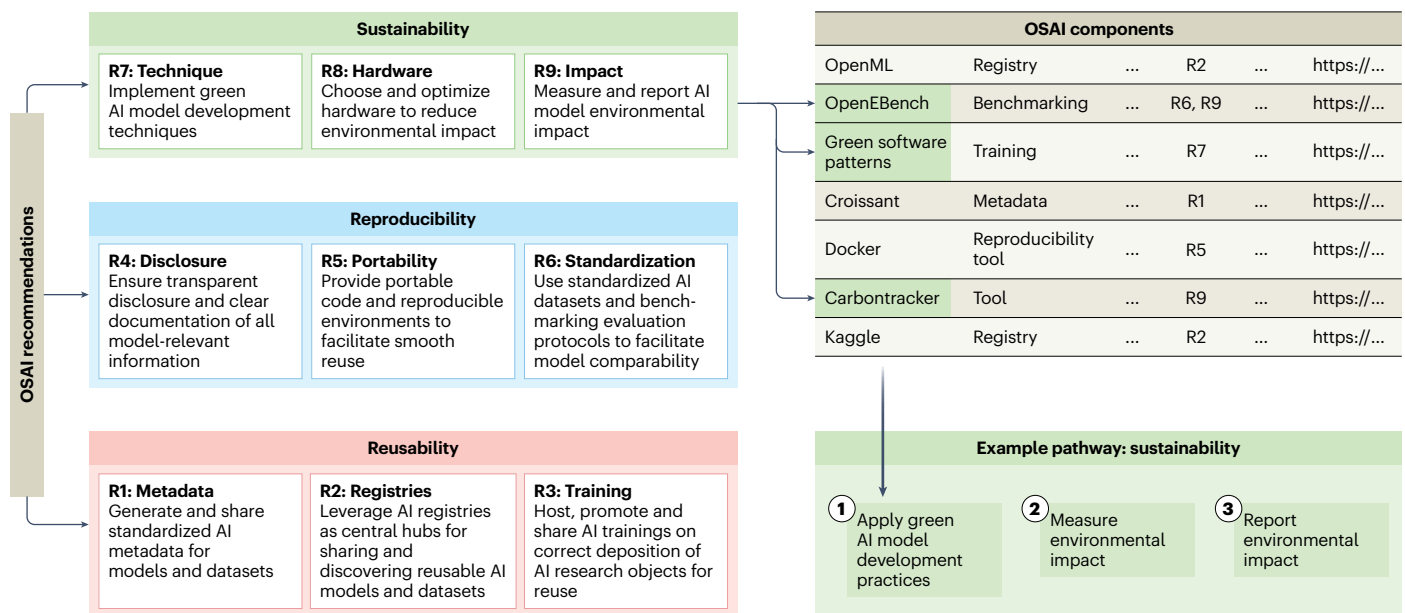


Fig. 2 | OSAI: recommendations and ecosystem components. OSAI recommendations covering nine practical and actionable solutions addressing practical issues across the three challenge areas of reusability, reproducibility and environmental sustainability. These recommendations are directly mapped to the OSAI ecosystem components to support their implementation. The OSAI ecosystem components mapped offer flexibility and inbuilt redundancy

with multiple solutions per recommendation across the full list. Both the OSAI recommendations and ecosystem components can further be chained into actionable pathways to best guide researchers to implement AI best practices, as shown with an example pathway covering environmentally sustainable model reporting. Created in BioRender; Farrell, G. <https://BioRender.com/xmwitsv> (2026).

HPC optimization support services (where accessible)—for example, POP CoE in Europe⁹⁴—can also assist with reducing AI workload environmental impacts.

R9: Measure and report AI model environmental impact. The life sciences community must begin to consistently measure and report AI model energy consumption to monitor their environmental impact. While no single metric is perfect due to measurement nuances and certain comparability limitations, researchers should aim to report on a set of minimal measurements such as total model development runtime (for example, CPU- or GPU-hours), hardware energy consumption (for example, kilowatt-hours) and operational carbon emissions (for example, CO₂ equivalent (CO₂e)). Planning the collection of these measurements in advance is crucial, preferably leveraging active measurement tools (for example, Carbontracker⁵⁵) or, where not possible, rough predictive calculators (for example, ML CO₂ Impact⁹⁵). Measurements of final model inference impacts are also valuable to help researchers consider choice of AI models if planning reuse. Reporting of measures should ideally be made available in a publication's methods, or hosted on life science AI model evaluation platforms (for example, OpenEBench⁸⁴), targeted community low-resource challenges such as MICCAI Fast and Low-resource (FLARE)⁹⁶ or dedicated leaderboards such as the Reinforcement Learning Energy Leaderboard⁹⁷.

OSAI: ecosystem mapping and pathways

While best practices and recommendations can be impactful, they can face barriers to adoption by researchers resulting from a lack of clarity on how exactly to implement them. This can lead to a reduction in their overall application across the intended community. Therefore, we have gone beyond an isolated set of recommendations by creating a dedicated digital resource for OSAI to support practical implementation (<https://osai.dome-ml.org/>). The interactive resource details the OSAI recommendations and contains a corresponding registry of over 300 relevant components of the AI ecosystem. With this list of components, we have created a non-prescriptive mapping, connecting the OSAI-relevant components that can help achieve our recommendations

(Fig. 2). The mapping has inbuilt redundancy as it hosts many solutions for achieving the same recommendation, which can avoid situations such as the FAIR principles for AI models by Ravi et al.⁹⁸, wherein a key component for implementation, DLHub, is no longer available for use.

We have designed the AI ecosystem components list to be maintainable by the community in an open GitHub repository (https://github.com/BioComputingUP/OSAI_ecosystem/)⁹⁹. This method is directly modeled after demonstrated community-supported resource lists, such as those within the Research Software Quality kit (RSQkit)¹⁰⁰. In this way, our landscaping builds upon the previous efforts shared in static publications—for example FAIR for AI by Huerta et al.⁴⁰. This collaborative approach will support longevity and contemporary relevance of the components captured, made possible through ongoing community maintenance encouraged by prominent contributor credit and recognition. Additionally, the GitHub has been enabled for Zenodo-supported version releases (<https://doi.org/10.5281/zenodo.17964970>) to ensure long-term persistence, availability and citability.

The AI ecosystem components and OSAI recommendations mapping are designed to be an openly accessible resource for linking researchers to the many possible components enabling OSAI research practice. The ecosystem components captured are not solely restricted to OSAI, and include important AI research platforms, tools and training, which expands its functionality to aid general AI research solution discoverability. We recognize the diversity across life science sub-domains and organizations, which may adopt varying AI guidance of narrower, wider or overlapping scope to those of our OSAI recommendations. Therefore, the captured AI ecosystem components are openly shared for interoperable reuse by other initiatives. For example, they could be repurposed for mapping to existing life science AI research guidance frameworks, or hosted on other AI platforms to aid connecting researchers to relevant supports.

Figure 2 highlights the OSAI recommendation mappings to the AI ecosystem components and how these can be further structured into guided OSAI implementation pathways. The OSAI web resource currently hosts three reference OSAI implementation pathways (<https://osai.dome-ml.org/>)

osai.dome-ml.org/pathways/), which cover reusability, reproducibility and environmental sustainability. They chain together the OSAI recommendations with a narrower subset of recommended components to actionably achieve OSAI in a step-by-step manner. An example OSAI implementation of the APNet model¹⁰¹ is also included on the pathways web page showcasing a direct user journey highlighting a lightweight implementation of the nine recommendations. In the long term, we envision wider community adaptations of our OSAI reference implementation pathways into sub-domain-specific pathways that highlight the most relevant AI ecosystem components in more targeted assemblies. We foresee toolkit resources modeled on the Turing Way⁷⁵, which are designed to support and disseminate community-cocreated research best practices, as resources that could be leveraged for hosting these pathways. Several existing toolkits addressing data^{12,102,103} or software¹⁰⁰ guidance could be considered for reuse to avoid proliferation of resources requiring large community and financial commitments to create, host and maintain them. We view additional guided pathways as the important next step toward supporting OSAI research practice across sub-domains and encourage global bodies working on mitigating challenges related to AI research come together to codevelop further pathways to guide AI best-practice implementations.

Emerging challenges and opportunities for OSAI

An emerging AI application in the life sciences is the use of AI agents and advanced agentic AI systems. AI agents can independently perform specific tasks of a narrow remit and can be further orchestrated into cohesive agentic AI systems to autonomously undertake diverse tasks with minimal human interaction¹⁹. Example AI agent applications include ontology maintenance (for example, Uberon¹⁰⁴) and spatial omics data analyses (for example, OmicsNavigator¹⁰⁵). More complex agentic AI systems like Biomni¹⁰⁶ have demonstrated the ability to perform biomedical research tasks ranging from generating wet-lab cloning protocols to analyzing heterogeneous datasets for rare disease diagnosis. The foreseen growth and uptake of these impressive agentic AI resources raise additional concerns for ensuring OSAI-compliant research methods.

Agentic AI systems introduce major OSAI concerns for both reproducibility¹⁰⁷ and environmental sustainability¹⁰⁸. Regarding reproducibility, these systems operate across dynamic data sources, tools and LLMs, which can lead to varied outcomes when tasks are executed across different periods due to underlying component changes such as LLM updates. In response, transparent documentation, thorough task execution logs, and open information sharing with an emphasis on explainable AI are vital to ensure interpretable and reproducible agentic systems. Regarding environmental sustainability, the increased computational demands of agentic AI are a key concern. These systems can run for longer and therefore use substantially more energy than traditional LLM prompting. Additionally, the common need for repeated runs when tasks fail further increases resource usage. Therefore, leveraging OSAI recommendations to mitigate these issues across the areas of reproducibility and environmental impact will be crucial. For example, there will be a growing need for transparent and comprehensive AI methods disclosure (OSAI R4) using checklist frameworks to support agent reproducibility. Furthermore, closer attention to the implementation of rene AI techniques (OSAI R7) and accountable AI resource use impact reporting (OSAI R9) will be crucial for monitoring and reducing negative agentic AI environmental impacts.

Another challenge for OSAI alignment is within the evolving frameworks of global policies like the EU AI Act¹⁰⁹, and the US NAIIR¹¹⁰ and related AI Action Plan¹¹¹. While such AI frameworks offer high-level policy on areas of responsible AI, most lack direct implementation guidance and are not yet closely connected to open science strategies to support reusable, reproducible and environmentally sustainable AI outputs. Addressing this gap at the policy level would enhance the uptake of AI best practices like OSAI in academia by feeding into

compliance mandates from funders and institutions during their grant proposal assessments and researcher assessments, respectively.

Currently, publications take priority in research assessment. To tackle this, global efforts like the Declaration on Research Assessment (DORA)¹¹² and the Coalition for Advancing Research Assessment (CoARA)¹¹³ are reforming research assessment practices to elevate the evaluation of research assets beyond publications. Such frameworks can help introduce researcher incentives to implement AI best practices by rewarding excellence of nontraditional research artifacts, such as formal contribution toward career progression and recognition during funding proposal decisions in support of research best-practice adherence.

Industry, in contrast, faces even starker barriers to adoption of open science and FAIR best practices. By design, companies must generate profit. They are, therefore, financially motivated and have less capacity or incentive to advance science for open exploitation in the same way as academia. This misalignment stemming from divergent incentives is a major concern for AI research asset outputs. Mitigation strategies such as stronger journal publication rules, and new policy and legislation could help to address this. Additionally, stipulations within academic collaborations to adhere to open science and FAIR principles could facilitate a closer alignment. This is important to tackle with the growing number of AI companies extending efforts into life science research, such as Google Deepmind's AlphaFold³ and Apple's SimpleFold¹¹⁴. A flagship example of these issues is the AlphaFold 3 (ref. 115) model, which was initially published without any available code. This led to heavy criticism¹¹⁶ and necessitated a response¹¹⁷ to address this.

Beyond these sectoral incentive challenges, a final cross-cutting consideration is the need to extend socio-ethical dimensions into AI research best-practice frameworks. While OSAI does not cover these areas, future extensions could leverage collaborative learnings from successful initiatives tackling socio-ethical concerns in the area of sensitive data, such as the GA4GH¹¹⁸, Bridge2AI⁸ and GDI Europe¹¹⁹, which have effectively incorporated experts on ethical, legal and societal implications in the design of their outputs. Possible extensions and cross-walks of this work (or other frameworks) could be undertaken through cross-domain international working groups (for example, via the RDA¹⁶) to facilitate this process. This will help to ensure not only are AI research outputs open and environmentally sustainable but also ethical, responsible and accountably designed.

Conclusion

Our OSAI recommendations offer practical support for researchers, developed with close consideration for both existing life science and domain-agnostic solutions (for example, from the RDA⁴¹, MLCommons²⁴, Green Software Foundation⁹²). They are designed to be complementary to comprehensive frameworks of wider AI scope, and directly focus on supporting research reform toward the production of OSAI research outputs. This work aims to foster a better understanding of OSAI practices and provides researchers with a navigable list of implementable solutions by consolidating the fragmented AI ecosystem components, and connecting them to actionable solutions.

While addressing the urgent challenges in areas of reusability, reproducibility and sustainability, we recognize the limitations of this work. We acknowledge a primary focus on AI models and less coverage of the important area of achieving AI-ready data, but instead refer to the existing efforts and guidance in this space, such as Bridge2AI⁸, the Open Data Institute's Framework for AI-ready data⁴³ and other supporting entries captured in our OSAI ecosystem. Furthermore, we do not directly address critical socio-ethical areas such as equitable, explainable and trustworthy AI, or the full breadth of responsible AI, given their complex, and variable requirements across regions. Instead we hope existing and future frameworks addressing these important areas can make use of our catalog of AI ecosystem components, which

is suitable to be repurposed for expanded resource mappings. We also recognize the initial landscaping of ecosystem components is not exhaustive, and will rely on open community contributions to ensure wider coverage, such as incorporating emerging AI model and dataset deposition trainings as they become available to better support our OSAI recommendation R3, 'Host, promote and share training and guidance on correct deposition of AI research objects for reuse'.

As a next step, the life sciences community must urgently fill the gap and collaboratively develop tailored AI best-practice implementation pathways. Our work lays the foundations through expanding the availability of practical OSAI recommendations supported by a diverse range of useful resources and provision of three reference implementation pathways. We foresee additional guided pathways as the necessary next step, potentially hosted in a new or existing toolkit-based resource, to assist life science AI researchers in a targeted manner. Ultimately, fostering a truly OSAI landscape necessitates a concerted effort from all stakeholders, including publishers, funders and policymakers. It will also be important to align community-driven efforts to the evolving policy landscape (for example, EU AI Act¹⁰⁹, NAIRR¹¹⁰), and next integrate with global networks such as FARR-RCN²⁵ or large-scale projects like AI4EOSC¹²⁰.

By embracing our nine OSAI recommendations, leveraging the mapped ecosystem of AI components and reference implementation pathways, researchers can usher in a shift toward reusable, reproducible and environmentally responsible AI becoming commonplace. This will ensure that life science AI outputs serve as reliable, and sustainable drivers for future life science breakthroughs serving society.

Code availability

All data and code are made available under a Creative Commons license CC BY 4.0: OSAI Ecosystem Interactive Website: <https://osai.dome-ml.org/ai-ecosystem/>; OSAI Ecosystem Data and Interactive Website Code - GitHub: https://github.com/BioComputingUP/OSAI_ecosystem/; GitHub—releases: https://github.com/BioComputingUP/OSAI_ecosystem/releases/; Zenodo—archived releases: <https://doi.org/10.5281/zenodo.15391273> (ref. 99); Software Heritage—archived releases: https://archive.softwareheritage.org/browse/directory/d1205b81b070e43af1c5d3e1493287518c5262d7/?origin_url=https://doi.org/10.5281/zenodo.15391273&path=BioComputingUP-OSAI_ecosystem-f7a6068&release=3&snapshot=d46b21f9a6057d0ea05675b2c2eb27745f93c9df.

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Author contributions

All authors contributed to the OSAI ecosystem and subsequent discussion as well as the initial draft of this manuscript. G.F. wrote

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Competing interests

R.F.-D. is an Industrial PhD candidate at University College Dublin with affiliation to IBM Research Dublin, who supports joint PhD research between both sites. L.C. is the director of SequenceAnalysis.co.uk. These affiliations are commercial in nature; however, the authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. All other authors declare no competing interests.

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